

Session 6.5

GRAM: A Retrieval System Built for One Literature — and What Agents Do to Research

Part five of five: rung three, how it was built and how the method transfers; then the close: the two dangers answered, the evidence on the researcher, and the economics of an agent

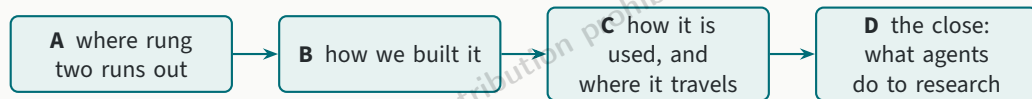


Zhigang Feng

This deck describes a system that is work in progress by Zhigang Feng, Jianjun Miao, Haoxuan Xie, Yaolong Zhong, and Yingli Zhou (2026). Its evaluation is a design; no result is reported here.

The Query This System Was Built For

“Life cycle with overlapping generations, idiosyncratic and aggregate shocks, incomplete intergenerational asset markets — which solver class is admissible for the joint stationary distribution in recursive equilibrium?”



- ▶ Every noun in it is a *condition*. The question is not what a paper is about; it is which method stays valid once those conditions hold together.
- ▶ A flat index matches the vocabulary and returns the famous paper that shares it — a different question, answered confidently.
- ▶ **Keep this sentence in view. Every design choice in this deck is an answer to it.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

The Pilot, and What These Pages Add

the newest frontier working papers

the last decade of journal papers

the works that fixed
mechanism and method

the founding early papers

one subfield: overlapping generations
and the life cycle

- ▶ **GRAM** — Generative Reasoning and Augmented Modeling — is a retrieval system for one literature in quantitative macroeconomics.
- ▶ The pilot **built 133 papers end to end**: the number carried through every stage into the index, not the number selected for it.
- ▶ The book already publishes the layered curation, the typed basis and the evaluation design; these pages add the reasoning and the engineering, and no result.
- ▶ **One subfield, deliberately. The claim is about a method, not about coverage.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.5; Feng, Miao, Xie, Zhong & Zhou (2026), work in progress; pilot build count.

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Where Rung Two Runs Out

A question about admissibility, and why a flat index answers it by prominence



A Question Flat Passages Answer Badly

a passage: "...aggregate shocks
...recursive equilibrium ..."

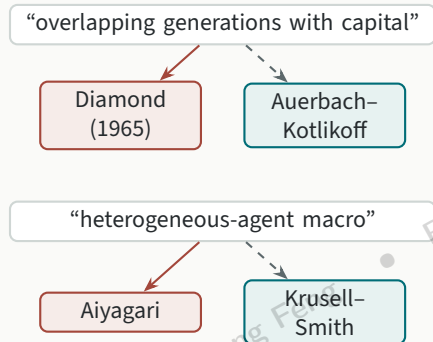


what it does not say: what
enters which constraint, and
what clears in which market

- ▶ Rung two's unit is prose: it carries a mechanism's words without the mechanism. Two papers that impose a borrowing constraint differently write nearly the same paragraph.
- ▶ A question about a *relationship* is not one about a topic. No passage holds the relation between two papers; it lives in the edge between them.
- ▶ **A flat index ranks on wording, and admissibility is not a property of wording.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

Wrong by Prominence



Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

- ▶ Embedding retrieval routes a query to the *most prominent* paper carrying its vocabulary, fitted to the asked-for conditions or not.
- ▶ In red, what comes back. In teal, the paper whose conditions actually match.
- ▶ **The vocabulary is present; the applicability filter is absent.**

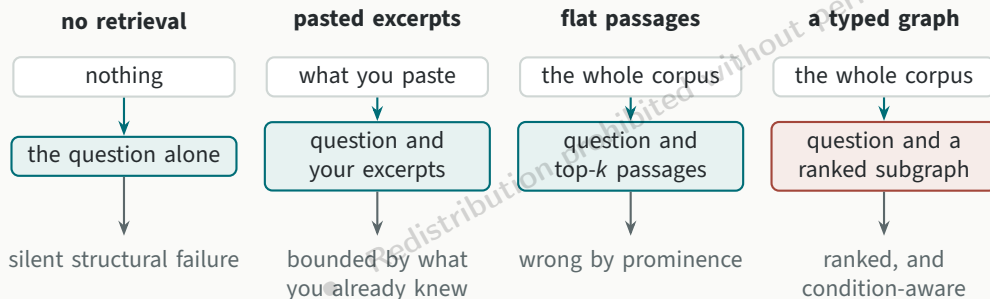
Admissibility Is Conditional

- ▶ Add aggregate risk to a life-cycle model and a solver class that was admissible stops being admissible; another one becomes so.
- ▶ The dependence runs from assumptions to methods, and neither paper's prose states it.
- ▶ **A vector of word co-occurrences cannot encode a dependency. A walk over typed objects can.**

Condition-aware retrieval returns a method *together with the conditions under which it holds*, rather than a passage that reads correctly and quietly drops its range of validity.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.5.

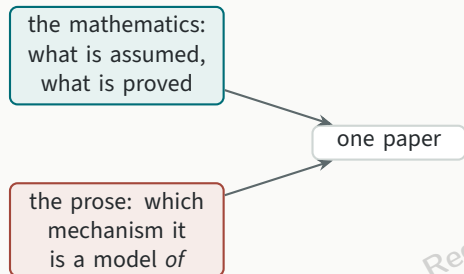
Four Ways to Give a Model Context, on One Question



- ▶ The generator is the same in all four. Only what reaches it changes.
- ▶ **These name each mechanism's failure mode on one kind of question. They are not a ranking of architectures.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

Why Economics Is Special: A Paper Is Mathematics *and* Prose



- ▶ Retrieval over the mathematics alone finds formally similar objects that model different things.
- ▶ Retrieval over the prose alone finds papers on the same topic under assumptions that do not hold.
- ▶ Economics needs both at once, which is why neither a mathematics search engine nor a general document index is the right starting point.
- ▶ **The index has to hold the structure, not a rendering of it.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

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How We Built It

Three models, a curated corpus, five stages, and a basis with names on its axes



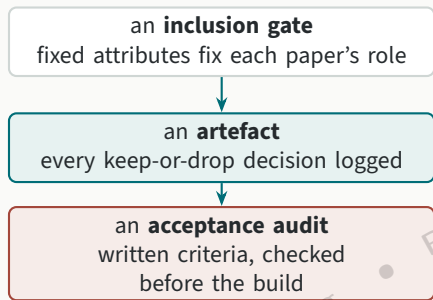
Three Models, Three Roles

role	modality	maps	what it is asked to do
reading	vision– language	page images to mark-up	rebuild the page: layout, reading order, tables, equations
reasoning	text to text	text to a typed record	resolve citations, and place the paper on the axes
embedding	text to vector	text to one space	put every element, and the question, in that space

- ▶ The role is named, not the product: each is replaceable, and the pipeline has to survive the replacement.
- ▶ **They never substitute for one another. A system that uses one model for every job cannot say which part failed.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

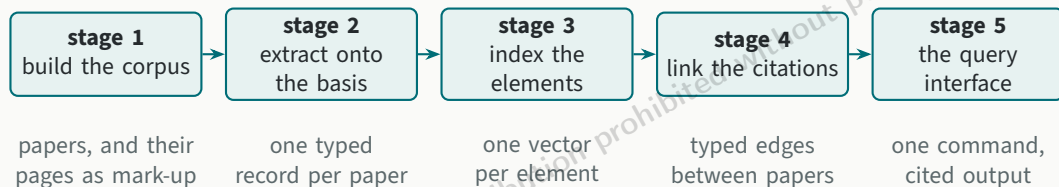
Corpus Construction Is Engineering: Tiers and Their Pressures



- ▶ A reading list is a personal judgement. A corpus is an artefact someone else has to be able to rebuild.
- ▶ Tiers settle three pressures, one layer at a time: coverage against focus, before against after the cut-off, and citation connectivity.
- ▶ **A retrieval system is only as good as the documents in its index, and that sentence is an instruction.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

The Five Build Stages



- ▶ No stage imports another: each reads the previous stage's files and writes its own, so any stage can be re-run alone.
- ▶ **Stages 1 and 2 are research; 3 to 5 are engineering. Keeping them apart is what lets you say which one failed.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

Why a Named Basis, and Not a Statistical One

named coordinates

β_1 β_2 β_3

each coordinate means something, so two papers compare one axis at a time

anonymous coordinates

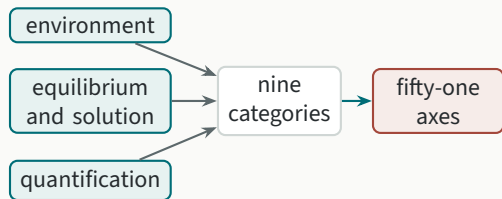
z_1 z_2 z_3

a direction that predicts well and explains nothing

- ▶ The analogy is a factor model: loadings on a few *named* factors,
 $R_i = \alpha_i + \beta_i^\top F + \varepsilon_i$.
- ▶ A dense embedding gives a paper 4096 coordinates — the principal components. The fifty-one named axes are the factors.
- ▶ **You can ask which axis two papers differ on. You cannot ask that of coordinate 2,317.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress; the analogy is to named-factor asset pricing.

The Basis: Three Blocks, Nine Categories, Fifty-One Axes



preferences, technology, market structure,
solution algorithm, welfare measure, ...

- ▶ three blocks, capitalised — the environment, equilibrium and solution, quantification — and beneath them nine categories and fifty-one axes.
- ▶ Seven typed edges say how the pieces relate: what enters an objective, what enters a constraint, what clears in a market, what derives what.
- ▶ **The schema is the research. Extraction is the mechanical pass you automate.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

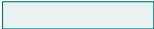
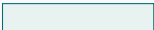
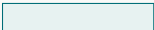
One Paper, One Provenanced Record

```
{ "axis": "borrowing_constraint",  
  "canonical_name":  
    "natural debt limit",  
  "description":  
    "the tightest limit that  
    never binds in autarky",  
  "source_lines": [412, 418],  
  "edges": [  
    { "type": "enters_constraint",  
      "target": "household_problem" } ] }
```

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

- ▶ One record per axis the paper touches, each carrying the lines it was read from.
- ▶ The canonical name is where papers merge — at the axis, never at paper-level similarity.
- ▶ **Provenance sits inside the retrieval unit, so every element returned can point at the sentence it came from.**

An Empty Axis Is Information

preferences	
technology	
market structure	
solution algorithm	

the blank row is a finding: this model has no such component ●

- ▶ A missing coordinate is a zero, not a hole in the data. It says the model contains no such ingredient.
- ▶ That is readable only because the basis is fixed in advance: with a schema invented per paper there is nothing to be absent from.
- ▶ **An absence you can query is a design choice. An absence you cannot see is a defect.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.5.

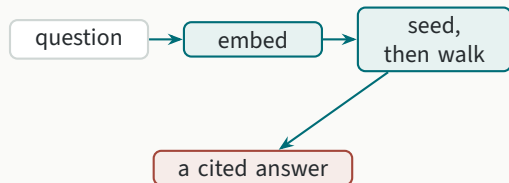
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How It Is Used, and Where It Travels

One command, a walk over a typed graph, and an evaluation that has not been run



At Query Time: Seed, Then Walk

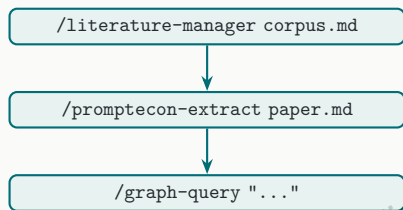


read-only and deterministic: the same question assembles the same context

- ▶ The model that indexed the elements also reads the question, so both live in one space.
- ▶ The nearest elements are not the answer; they seed a walk over the typed graph.
- ▶ The context policy is three numbers: eight seeds, then six element and four citation chunks.
- ▶ **The answer is written from the assembled block alone: cite it, or decline.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

The Interface Is a Slash Command



a verified manifest: what is in the corpus, and where each file came from

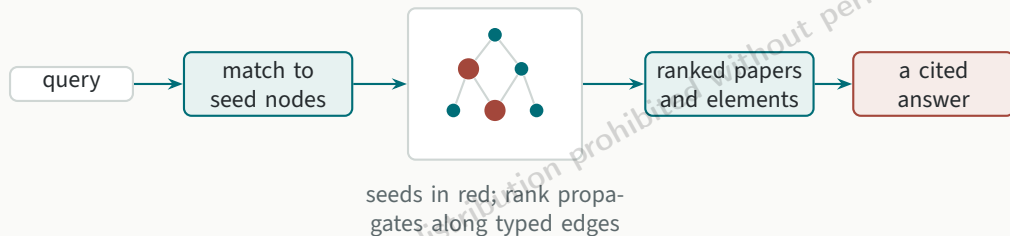
one typed record for that paper, placed on the axes

ranked papers, elements and citations — cited, or a refusal

- ▶ A skill is a folder the model opens when it is needed (Deck 6.2). These are three of them, in the order a researcher uses them.
- ▶ **The interface is part of the deliverable. A retrieval system you have to remember how to invoke is one nobody uses.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

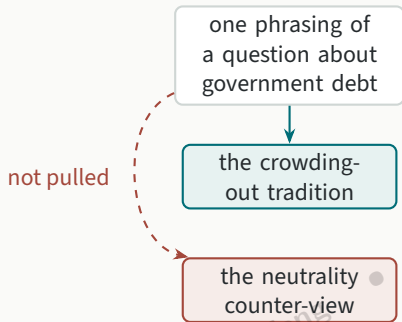
Graph Retrieval: Seeds, Propagation, Ranked Papers



- ▶ Far from the seeds, a paper never reaches the model: filtering precedes generation.
- ▶ End to end: embed, seed, propagate, assemble ten chunks, answer or decline.
- ▶ **A walk can express “reachable from these conditions”. A nearest-neighbour lookup cannot.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

Retrieval Honesty: The Counter-View That Did Not Surface

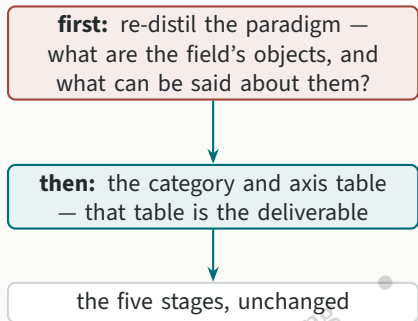


- ▶ In one demonstration the Ricardian-equivalence counter-view never surfaced, although the paper making it sits in the corpus.
- ▶ An answer is grounded in whatever retrieval pulled, so it is only as balanced as the phrasing that pulled it.

The assembled context must be printable. One-sidedness is visible there and nowhere else — never in the prose of the answer.

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress; a demonstration, not a measurement.

Reusing the Recipe in Another Field



- ▶ The engineering transfers unchanged. The basis does not: fifty-one axes for structural macroeconomics say nothing about an empirical literature.
- ▶ Re-deriving the paradigm is months of reading, and it is the step people try hardest to automate.
- ▶ **The table of categories and axes is the contribution. The pipeline around it is plumbing.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

From Fifty-One Axes to Your Field: A Worked Transfer

object	category	axes
identification strategy	the research design	variation; comparison group; timing
data	source and unit	survey or register; unit; years
population	treated, and compared	age band; income band; jurisdiction
estimand	the parameter	average or local effect; which margin
outcome	what is measured	coverage; utilisation; health; finance

- ▶ The previous frame's three steps, run once on the empirical literature on health-insurance expansions: five objects, none a preference or a technology.
- ▶ Each object becomes a category; its axes are the values a paper can take, fixed before reading, so an empty axis still reads as information.
- ▶ **Stages and models unchanged. The table is the transfer, and the economist writes it.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.5; Feng, Miao, Xie, Zhong & Zhou (2026), work in progress; axes illustrative.

The Evaluation Is a Design. Nothing Is Reported Yet.

1 the model alone, no retrieval

?

2 flat passages, one
dense bilingual embedder

?

3 typed axes and
a citation graph

?

one generator across all three: the
manipulated variable is retrieval

- ▶ **Condition 2 is the system Deck 6.4 built** — the control condition rung three is measured against.
- ▶ Three questions: is the extraction faithful; which component earns any gain; does typed retrieval beat flat passages.
- ▶ Two tasks: an expert panel, and generating an algorithm and its code.
- ▶ **No condition has been scored. Nothing in this deck is a result.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.5; Feng, Miao, Xie, Zhong & Zhou (2026), work in progress — design only.

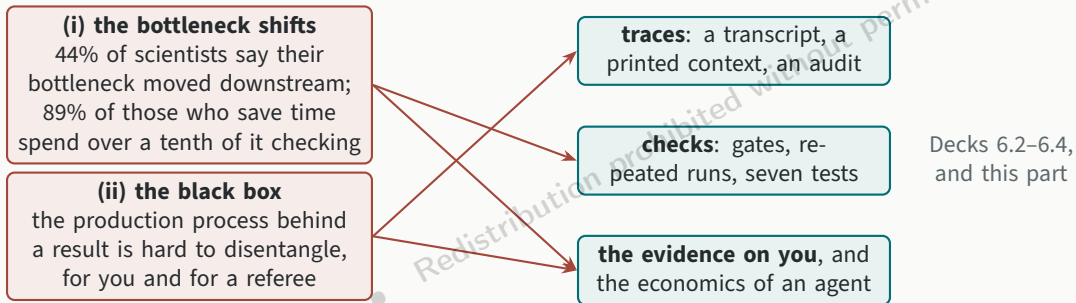
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Closing: What Agents Do to Research, and to You

The two dangers answered, the evidence on the researcher, and the economics of an agent



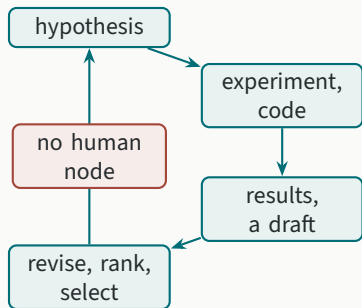
Two Dangers, Answered



- ▶ Deck 6.1 named both dangers and the survey behind (i). The answer to both is a contract kept in files: spec, trace, gate.
- ▶ **This is the part where the session stops adding capability and starts pricing it.**

Source: Codreanu, Imas, Mateos-Garcia et al. (2026), *AI in Science: Early Insights*, pp. 5, 25–26 (read 23 Sep 2026); Deck 6.1's closing frames; Holmström (1979).

Closing the Loop, and Where It Stops

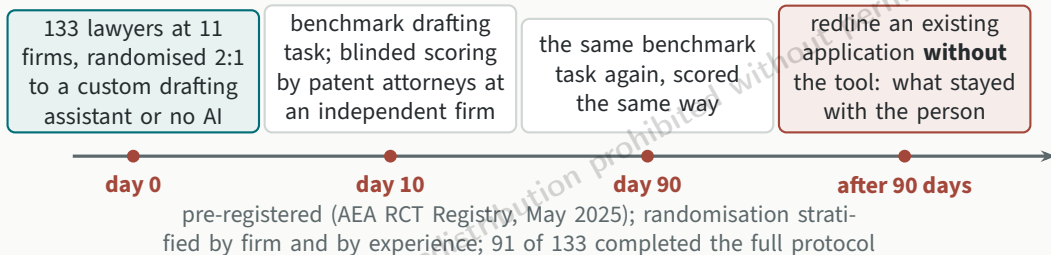


converges where the
criterion is auto-decidable
and iteration is cheap

- ▶ Lu et al. (2026, *Nature*): an end-to-end system's workshop paper scored 6, 7 and 6, above the bar, and was withdrawn as agreed; its failures included hallucinated citations and wrong implementations.
- ▶ Where a published table decides, loops converge first: the best agents match a reported sign over 85% of the time (Kohler et al., 2026).
- ▶ Scheduled, unattended runs remove the turn in which you would have noticed.
- ▶ **Nothing inside the loop knows whether the question was worth asking.**

Source: Lu et al. (2026), *Nature* 651: 914–919, 25 Mar 2026; Kohler et al. (2026), arXiv:2604.21965 v1, preprint; Anthropic, routines in Claude Code, 14 Apr 2026; read 23 Sep 2026.

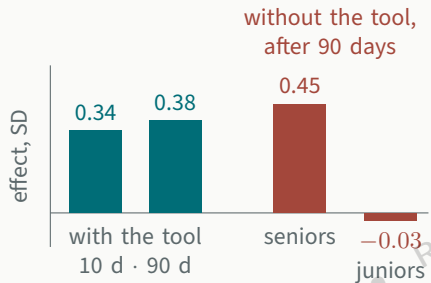
Does AI Assistance Enhance or Erode Expertise? The Design



- ▶ A three-month field experiment in patent drafting, at practising firms, with a rubric covering enforceability, accuracy, strategic ambiguity, completeness and clarity.
- ▶ **The design answers two questions at once: what the tool adds while you use it, and what remains when it is taken away.**

Source: Autor, Rodchenko, Martin, Iscenko, Strand, Pearl & Ferere (2026), NBER WP 35720, Sept 2026, pp. 1–3 (read 23 Sep 2026; not peer-reviewed).

Seniors Kept the Gain; Juniors Split



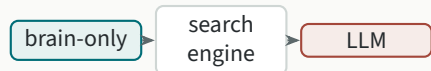
“Foundational expertise may be a prerequisite for extracting durable skill from AI-assisted practice.”

- ▶ With the tool: +0.34 SD at 10 days, +0.38 SD at 90, larger for juniors. Without it: +0.32 SD overall, all of it seniors (+0.45 SD); juniors -0.03 on average and *bifurcated* — fewer mediocre scores, more poor ones, more good ones.
- ▶ The counterpoint to Deck 5.5’s “cognitive levelling” (Noy & Zhang; Brynjolfsson, Li & Raymond): it holds only while the tool is in hand.
- ▶ **The largest gains accrued to the lawyers who retained the least.**

Source: Autor et al. (2026), NBER WP 35720, pp. 2, 18, 23–24, 38 (read 23 Sep 2026); effect sizes in SD, not peer-reviewed.

What Automation Does to the Researcher: Cognitive Debt, Hedged

sessions 1–3: connectivity,
strongest to weakest



“cognitive activity scaled down
in relation to external tool use”

session 4:
LLM →
brain-only

session 4:
brain-only
→ LLM

the crossover: LLM-to-brain showed
reduced alpha and beta connectivity

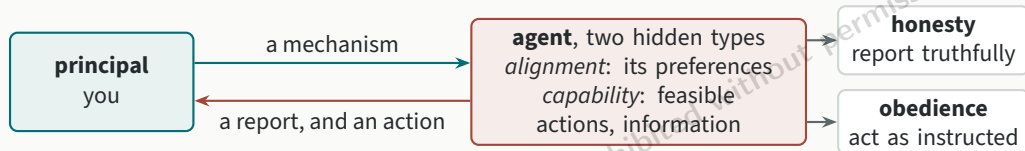
reliable tools get checked less

Preprint, under review (arXiv v2, 31 Dec 2025); $n = 54$, 18 in session 4. Gerlich (2025): AI-tool use and critical thinking, $r = -0.68$, mediated by cognitive offloading — a correlation in a UK survey ($n = 666$), which the book hedges the same way. Neither is a causal estimate.

- ▶ In session 1, 83.3% of the LLM group (15 of 18) could not quote a sentence of the essay just handed in; by session 2 the share had fallen.
- ▶ **Thin evidence, consistent direction: what you hand over, you stop rehearsing.**

Source: Kosmyna et al. (2025), arXiv:2506.08872 v2, pp. 31, 36; Gerlich (2025), *Societies* 15(1):6; Parasuraman & Manzey (2010); Miao & Feng, ch. 15, §4.4.3; read 23 Sep 2026.

Mechanism Design When Alignment and Capability Are Both Unknown



one-sided imitation: “capabilities can be concealed but not counterfeited” $\Rightarrow$ a revelation principle, and implementable policies characterised by nested cyclical monotonicity

- ▶ Deck 1.1’s principal could not observe effort; this agent hides what it wants and what it can do, so a mechanism “must incentivize both honesty and obedience”.
- ▶ A weak agent cannot fake strength, a strong one can play weak: this asymmetry makes it tractable; eliciting higher-order beliefs can discipline several agents at once.
- ▶ **Deck 6.1’s two dangers in one contract: unseen effort, unseen ability.**

Source: Bergemann, Koh & Morris (2026), “Mechanism Design for Alignment and Control”, arXiv:2609.01595 v1, 1 Sep 2026, abstract (body not read); Deck 1.1’s Holmström frame.

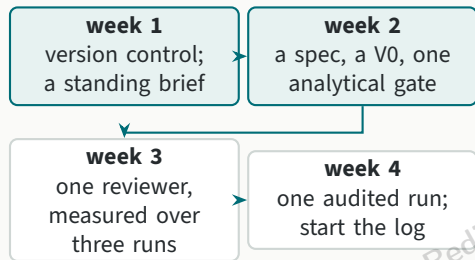
Five Applications, Read as Team-Design Questions

application, per the abstract	the team-design question it becomes
(i) sandbagging: a more capable agent pretends to be less capable	the sub-agent that reports “not feasible”: hiding capacity, or lacking it? Probe it on a task you know it can do
(ii) an alignment–interpretability trade-off: substitutes in the instrument, complements in value	danger (ii): a reviewer whose reasoning you can read is worth more than a stronger one you cannot
(iii) discipline via peer scoring	the verifier sub-agent (Deck 6.3): a second agent scores the first, on a rubric written first
(iv) coupling rewards to induce competition among agents	two solvers on one gate, one comparison; neither sees the other’s answer
(v) scalable oversight and reward shaping	the gate itself — and what a loop does once it is told what will be measured (Deck 1.1’s Lucas critique)

► The five stylised examples are Deck 6.3’s instruments, stated as a theory.

Source: Bergemann, Koh & Morris (2026), arXiv:2609.01595 v1, 1 Sep 2026, abstract; the right column is this course’s reading; Deck 1.1’s Holmström and Lucas-critique frames.

Your First Thirty Days, and the Job Description



the furthest point you can still
check is the furthest you should go

- ▶ Four weeks, one habit each, on a project you already have. Read first: Korinek (2025), NBER 34202, the economist's own manual, an agent built step by step; Galiani, López & Sosa (2026), NBER 35588, what agency bought on econometric tasks.
- ▶ You remain the **principal investigator**, and the **project manager** of Deck 6.1: gate every good, assign it, cross-verify it.
- ▶ **Agents lay bricks quickly. Remain the client of a prior other than your own.**

Source: Korinek (2025), NBER WP 34202; Galiani, López & Sosa (2026), NBER WP 35588; Parasuraman & Manzey, *Human Factors* (2010); Miao & Feng, *AI for Economists*, ch. 15, §5.

What the Course Promised, and Where It Was Built

promised in Deck 1.1

text measures and a small transformer

a grounded agent that leaves a trace

citations verified

numerical outputs verified

failure paths verified

reproduction from a clean, offline start

discharged

Session 5's lab, end to end

block A: 7 readable turns

block D: each sentence carries its meeting

block B: 96 numbers recomputed

block D's three failures; block F's gate

fixed seeds, no network, hashes checked

- ▶ The course opened on a sentence: fluent output is not a result — a result survives a check it could have failed. Block B is that sentence with numbers in it.
- ▶ **So demand four things of anything an agent hands you: the trace that made it, the numbers recomputed from the source, the claim stated apart from the arithmetic, and the rule that would have stopped it. Short of all four, it is a draft.**

Source: Deck 1.1, what students build and verify; this session's lab, blocks A to F.